

IBM DATA SCIENCE CAPSTONE

SpaceX Falcon 9 Launch Outcomes

Technical presentation and analytical report

GitHub: <https://github.com/itsilger97-hub/IBM-SpaceX-Capstone>

Dataset: historical IBM course CSV • 101 rows × 10 columns

Observed outputs and provisional metrics are distinguished; unverified application screenshots are not fabricated.

1. Executive Summary

- Objective: analyze historical Falcon 9 launch records and build a baseline classifier for landing outcomes.
- Dataset: 101 rows and 10 fields from the IBM course CSV. A Colab API request returned HTTP 525, so the historical CSV was used.
- Workflow: data checks, landing-outcome EDA, SQL aggregation queries, Folium map, Plotly chart and four baseline classifiers.
- Initial accuracy: Logistic Regression 76.19%, SVM (RBF) 76.19%, Decision Tree 76.19%, Random Forest 71.43%. These scores remain provisional until retraining with the explicit target.

2. Introduction & Business Question

- Question: do payload mass, orbit, launch site and booster version help classify successful landing outcomes?
- Falcon 9 first-stage recovery is relevant to reusability economics. This is an educational analysis, not a flight-safety tool.
- Positive class: Success, Success (drone ship), Success (ground pad); other labels are the comparison class.

3. Data Collection & Wrangling Methodology

- One API request returned HTTP 525 in Colab. A separate endpoint returned HTTP 200 for 100 recent records and four columns; it was kept separate from the historical analysis.
- Historical course data: 101 rows × 10 columns. Fields include Date, Time (UTC), Booster_Version, Launch_Site, Payload, PAYLOAD_MASS__KG_, Orbit, Customer, Mission_Outcome and Landing_Outcome.
- Inspected shape, columns, types, missing values and duplicates; converted payload mass to numeric for visualization.
- Explicit target distribution: 61 successful-category records and 40 other outcomes.

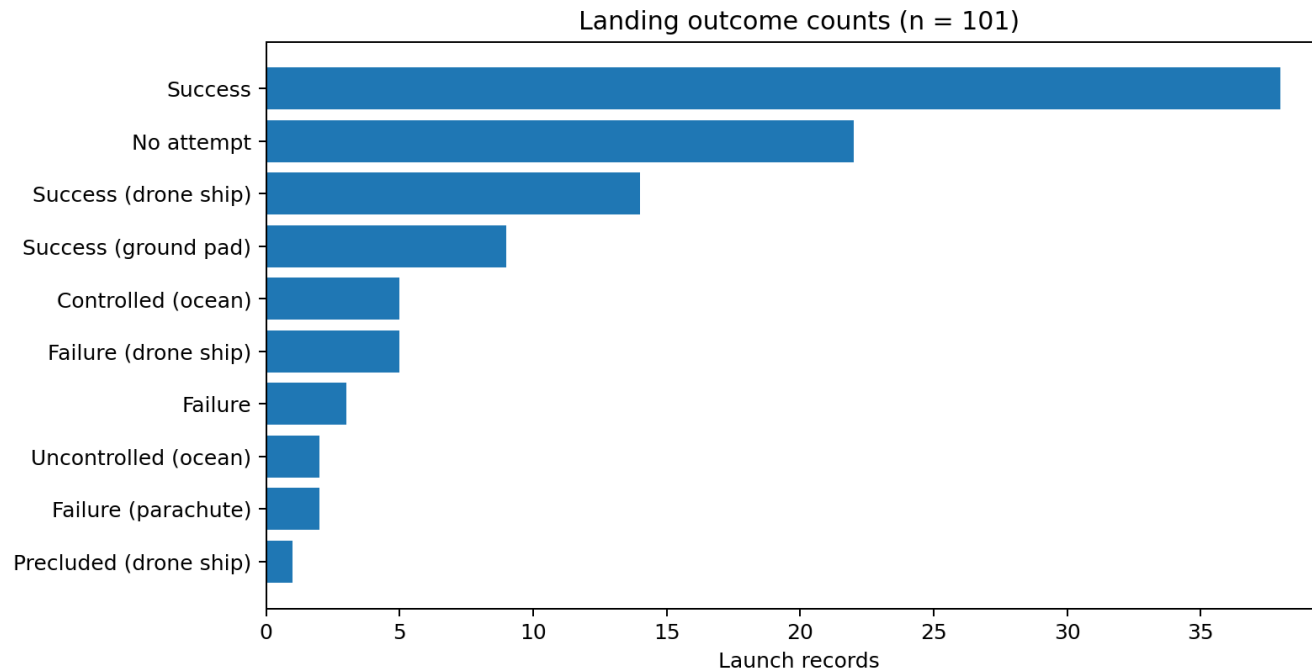
Dataset loaded successfully!
Rows: 101
Columns: 10

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-				CCAFS LC-	SpaceX		LEO	NASA		

4. EDA & Interactive Visual Analytics Methodology

- Count plots summarize landing outcomes, mission outcomes and launch-site frequencies.
- Payload histogram examines mass distribution and potential outliers after numeric conversion.
- SQL aggregation provides repeatable counts and payload range; Folium provides facility markers; Plotly Express visualizes outcome counts.
- These are descriptive analyses; they do not establish causal relationships.

5. EDA Results: Landing Outcomes



Counts: Success 38; No attempt 22 (displayed as 21 and 1); Success (drone ship) 14; Success (ground pad) 9; Controlled (ocean) 5; Failure (drone ship) 5; Failure 3; Uncontrolled (ocean) 2; Failure (parachute) 2; Precluded (drone ship) 1.

```
Landing_Outcome
Success          38
No attempt       21
Success (drone ship) 14
Success (ground pad) 9
Controlled (ocean) 5
Failure (drone ship) 5
Failure          3
Uncontrolled (ocean) 2
Failure (parachute) 2
Precluded (drone ship) 1
No attempt       1
Name: count, dtype: int64
```

6. EDA with SQL: Queries & Observed Results

The dataframe was loaded into SQLite as SPACEXTBL. The landing outcome count table below is based on the counts supplied from the notebook; exact launch-site totals and payload extrema were not available in the shared output.

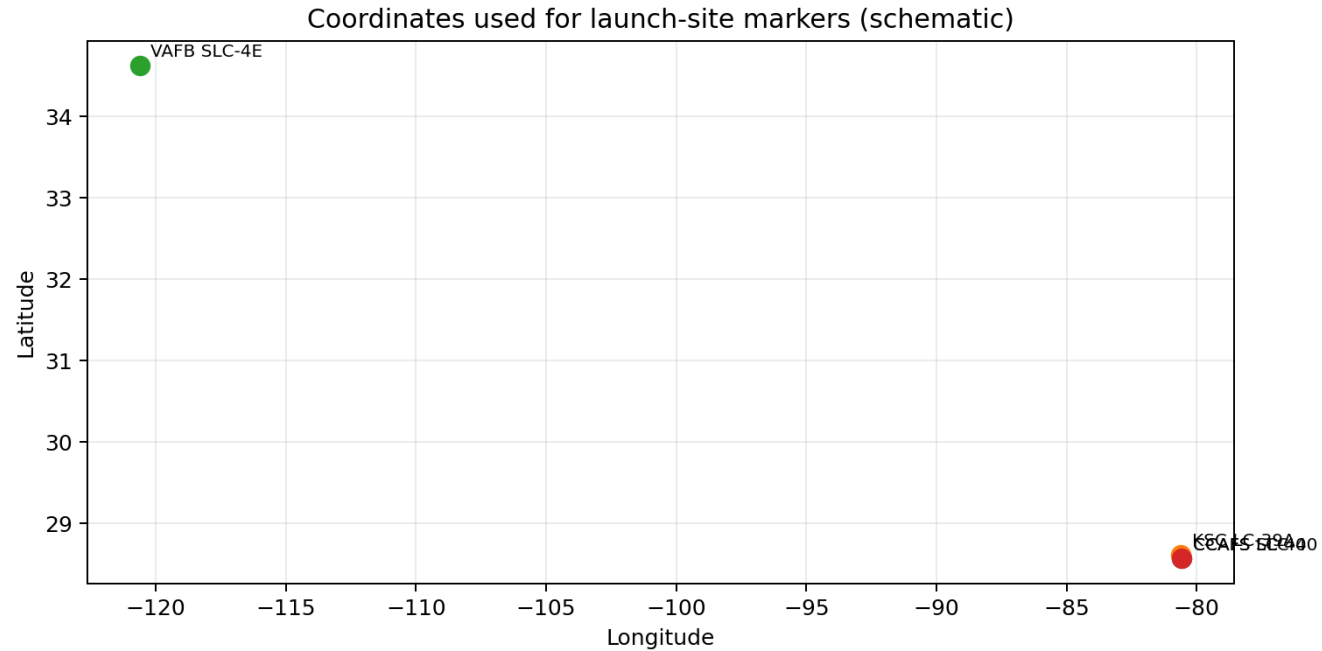
Landing outcome	Count
Success	38
No attempt	22
Success (drone ship)	14
Success (ground pad)	9
Controlled (ocean)	5
Failure (drone ship)	5
Failure	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

```
SELECT Launch_Site, COUNT(*) AS Total_Launches FROM SPACEXTBL GROUP BY Launch_Site ORDER BY Total_Launches DESC;
```

```
SELECT MIN(PAYLOAD_MASS_KG_) AS Min_Mass, MAX(PAYLOAD_MASS_KG_) AS Max_Mass FROM SPACEXTBL;
```

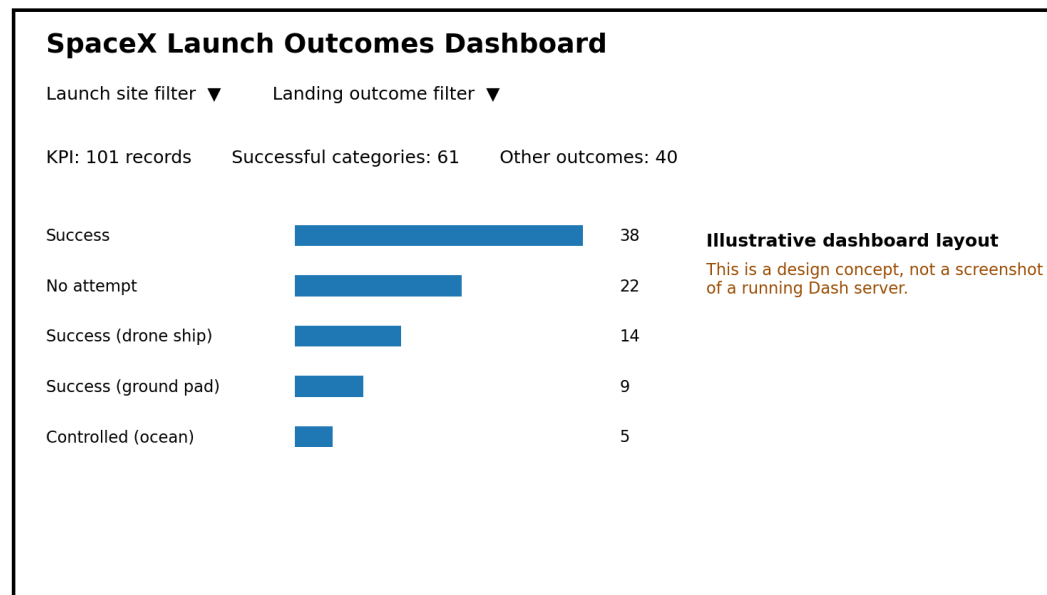
```
SELECT Launch_Site, Mission_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTBL GROUP BY Launch_Site, Mission_Outcome;
```

7. Folium Map Results



- Facility labels used: CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E and a nearby CCAFS LC-40 label.
- The coordinate plot is a static representation; a Folium HTML map is included when the library is available.
- Verify closely spaced or inconsistent CCAFS labels before interpreting them as distinct facilities.

8. Plotly Dash Dashboard Results

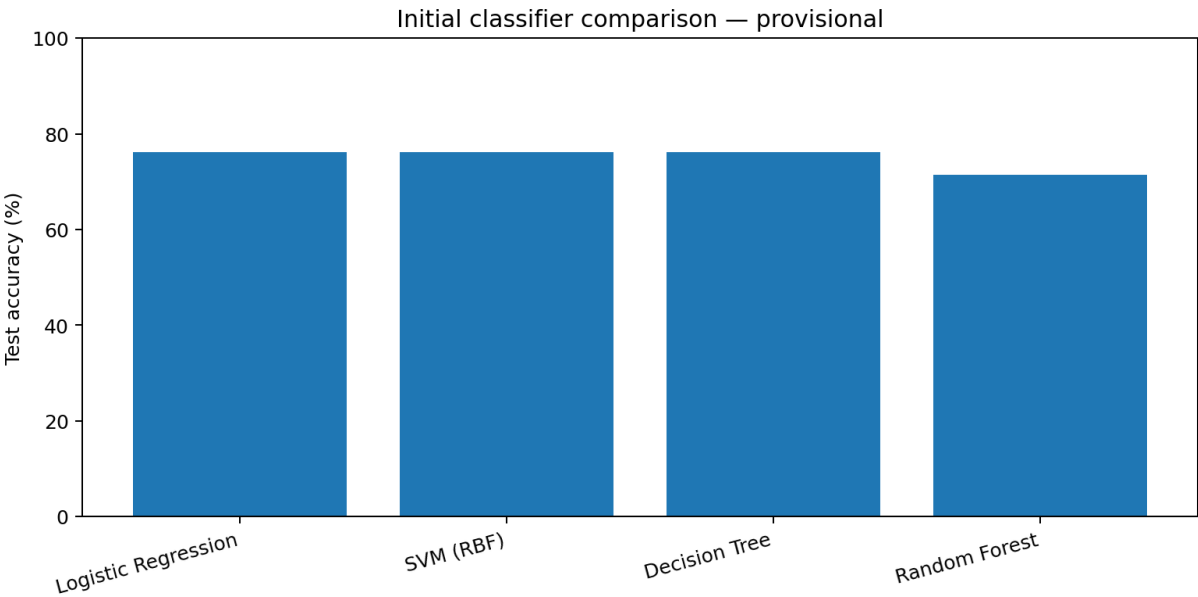


- The Plotly Express outcome chart ran in Colab; hover interactions expose category values.
- The included Dash source implements launch-site and landing-outcome filters with a callback. The illustration above is a concept, not a screenshot of a running server.

9. Predictive Analysis Methodology

- Target: 1 for the three explicit successful landing labels, 0 for other observed outcomes.
- Features: PAYLOAD_MASS_KG_, Orbit, Launch_Site, Booster_Version.
- Numeric preprocessing: median imputation and scaling. Categorical preprocessing: most-frequent imputation and one-hot encoding.
- Stratified 80/20 train/test split, random_state=42; Logistic Regression, SVM (RBF), Decision Tree, Random Forest.

10. Predictive Analysis Results



Model	Initial accuracy
Logistic Regression	76.19%
SVM (RBF)	76.19%
Decision Tree	76.19%
Random Forest	71.43%

Provisional: these are initial notebook values; the target definition was explicitly corrected afterward, and updated metrics were not shared.

11. Model Evaluation & Limitations

- Three classifiers tied at 76.19%; accuracy alone does not select a unique winner.
- With 101 rows, the held-out test set is small and a few predictions can materially change accuracy.
- Final evaluation should include confusion matrix, precision, recall, F1-score and class support after corrected-target retraining.
- Do not present initial metrics as final until the pipeline is rerun.

12. Innovative Insights

- Normalize repeated labels before aggregation; duplicate 'No attempt' entries can split one category.
- Keep 'No attempt' distinct from an actual failed recovery.
- Compare outcomes by orbit, site, payload and booster version, treating counts as descriptive associations.
- Use class-specific precision and recall rather than accuracy alone.

13. Conclusion

- The workflow integrates data collection, wrangling, EDA, SQL, spatial visualization, interactive charts and classification.
- The course sample contains 101 records; the explicit target contains 61 successful-category records and 40 other outcomes.
- Initial accuracy peaked at 76.19% for three models, but final model selection requires retraining with the explicit target.
- Evidence still to capture: SQL output tables, live Folium/Dash screenshots and final model metrics.

14. GitHub & Deliverables

Repository: <https://github.com/itsilger97-hub/IBM-SpaceX-Capstone>

- Package contains README, notebook, analysis Python script, SQL queries, Dash source, charts and available screenshots.
- Upload the notebooks, Python files, CSV, SQL, screenshots, README, PPTX and PDF to the public repository.
- Run the included Dash app to capture the real dashboard screenshot; replace provisional model metrics after retraining.

Rubric item	Evidence in report
Executive summary / introduction	Detailed sections
Data collection / wrangling	Schema, API issue, checks, target
EDA visualization	Actual outcome counts and chart
SQL	Executable queries and observed outcome counts
Folium	Coordinate view + HTML map where available
Plotly Dash	App source + layout concept; live screenshot still needed
Classification	Methodology and initial accuracy table; provisional

Rubric item	Evidence in report
Conclusion / creativity / insights	Dedicated sections